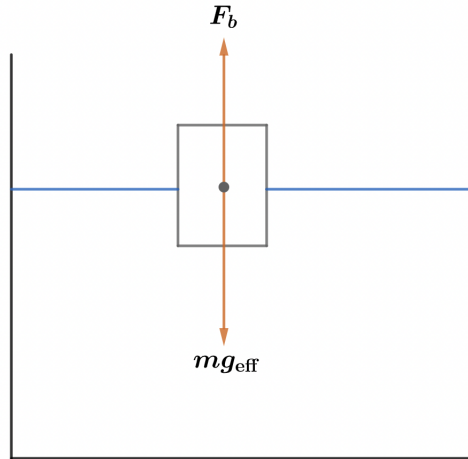


2022A F=ma Exam: Problem 21

Kevin S. Huang



The buoyant force (weight of water displaced) balances the weight of the cork in the inertial frames when the elevator moves at constant velocity. In the accelerating frames, all weights get modified by the same factor $mg \rightarrow mg_{\text{eff}} = m(g + a)$ so the forces are still balanced. The height of the cork doesn't change so the answer is E.